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Abstract

We present a case study of productive flyby users (PFB users) on a recommendation website. These users exhibit counterintuitive behavior: they input a large amount of data during their first visit but never return. This phenomenon can have both positive and negative impacts on the system. On the positive side, their high productivity contributes a substantial amount of data. On the negative side, they may input inappropriate ratings that violate the assumptions of recommendation algorithms, potentially undermining system performance. To better understand the nature and causes of this behavior, we investigated their motivations, expectations, reasons for leaving, and the potential risks associated with their ratings using a mixed-methods approach. Specifically, we conducted interviews with 11 users, surveyed 41 users, and analyzed the impact of 1,000 PFB users on the performance of recommendation algorithms for regular users. Our findings revealed diverse motivations among PFB users. Some engaged with the system merely to pass the time, while others had unrealistic expectations of the recommender system. Regarding rating quality, 27% of surveyed users admitted to rating movies they had not seen, citing reasons such as browsing too quickly or attempting to manipulate the algorithm. Notably, users who reported leaving because they were "just killing time and forgot about the website" were the most likely to rate unseen movies. Overall, PFB users significantly influence recommendation algorithms and their performance for regular users. While some subgroups negatively affect prediction accuracy, others provide valuable data contributions. We discuss strategies for recommendation websites to better support these users or mitigate the impact of inappropriate ratings.

CCS Concepts

• **Human-centered computing → Empirical studies in collaborative and social computing**.

Keywords

Recommender System, User Behavior Understanding, Data Integrity

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1 Introduction

User retention and behavior understanding are critical to the success of many recommender systems. Prior studies have investigated various algorithms, including reinforcement learning [8], contrastive learning [14], and feature-engineered XGBoost [29], to model user behaviors and mitigate user churn. However, the reasons users initially engage with these systems and the interpretation of their churning behavior across different usage patterns remain open research questions.

Inspired by our observations on a non-commercial personalized movie recommender system MovieLens[1], we identified a unique group of users exhibiting an intriguing behavior pattern. These users are **productive**, contributing substantial amounts of data during their **first visit** (within 24 hours of their initial login). However, they then **fly by**—never returning to MovieLens after that initial interaction. Notably, 48% of new users who input over 100 ratings during their first visit fall into this category. We define these

[1] https://movielens.org/

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highly invested yet transient users as **productive flyby users** (PFB users).

PFB users have been overlooked in prior studies on newcomers to recommender systems. Many studies have found that the level of investment new users make during their first visit is positively correlated with their likelihood of returning [15, 27, 37]. This observation contrasts with the behavior of PFB users, who invest significantly during their initial visit but still leave the system permanently. Focusing on effort motivations, reasons for leaving, and risks related to rating integrity, we aim to address the following three research questions:

> ***RQ1 (Experience)***: What are PFB users' motivations for visiting the recommender, expectations for contributing ratings, and reasons for leaving the platform?
> ***RQ2 (Risk)***: What rating integrity risks can we identify from PFB users' ratings? And what are the reasons behind their inappropriate contribution?
> ***RQ3 (Impact)***: How can PFB users' inappropriate ratings negatively affect other regular users?

Our methods consisted of three activities to address the research questions. First, we identified 449 PFB users in our system through log analysis and distributed a survey to gather subjective insights about their experiences and rating contributions. From the survey respondents, we conducted in-depth interviews with 11 PFB users to further investigate their motivations, expectations, and reasons for leaving. During these interviews, we also sought to uncover the underlying causes of their inappropriate ratings. Finally, we performed an offline evaluation with 1,000 PFB users, examining the impact of their inappropriate ratings on the prediction accuracy for a sample of regular users within the system.

We outline the main contributions of this paper as follows:

- We identified three major motivations and expectations of PFB users: 1) *"Just-forgot"* users, who randomly drop in to kill time and forget about the platform; 2) *"High-expectation"* users, who arrive with unrealistic expectations and leave when the system fails to meet them; and 3) *"One-time"* users, who never intended to stay beyond their first visit.
- We found that 27% of PFB users exhibited rating integrity issues (e.g., rated movies they had never watched), with *"Just-forgot"* users being the most likely to rate unseen movies. The primary reasons for these inappropriate ratings included misinputs caused by rapid interaction and attempts to manipulate the algorithm.
- Our evaluation revealed that, overall, ratings from PFB users did not significantly affect the prediction performance of collaborative-filtering-based recommenders for regular users. However, inappropriate ratings from certain high-risk subgroups of PFB users did negatively impact recommendation accuracy for regular users.
- To the best of our knowledge, this is the first study to comprehensively analyze the entire user behavior flow of PFB users, from their motivations and expectations to their rating integrity in recommender systems. We provide in-depth discussions of the implications of this study and propose actionable recommendations for designing more robust recommender systems that deliver improved user experiences across diverse user groups.

In the rest of this paper, we first review relevant literature, then share details of the study procedure, followed by presenting the results of RQs and analysis with both statistical and qualitative data from the survey and interview data. Finally, we synthesize the findings and design implications for future user-centered recommender systems.

2 Related Work

2.1 New User Retention in Online Platforms

New user retention is a significant challenge for online communities and peer production platforms. For instance, approximately 60% of Wikipedia's new users never return to edit after their first session [22, 27]. Similarly, 54% of new users on Yahoo! Answers [15] and 50% of new users in Usenet groups [3] fail to return. Recommender systems face comparable challenges in providing personalized recommendations and delivering a positive user experience to new users, often referred to as the *cold start* problem [40]. Previous studies have explored the performance of various algorithms in enhancing user experience [23, 28] and improving preference prediction [30, 39].

Understanding new users' motivations, expectations, and experiences is considered a crucial step in studies aimed at retaining new users. Cooper et al. proposed that preventing user churn requires an onboarding process that effectively communicates a product's concepts and scope in a timely manner [11]. Similarly, Cascaes Cardoso identified the "purpose statement" as a foundational element of user onboarding, defining it as the process of understanding new users' goals and conveying the platform's relevance to their expectations [9]. Further, Badali et al. demonstrated that user motivation in MOOCs significantly influences retention rates [4]. Additionally, Ciampaglia and Taraborelli found that enabling Wikipedia's new users to provide feedback on their onboarding experiences led to increased retention rates [10]. However, little research has focused specifically on the motivations and experiences of productive flyby users.

New users' level of investment is often identified as a key factor in predicting their retention. For example, Karumur et al. observed that Wikipedia users who returned made an average of 15 edits on their first day, compared to only 3 edits by users who churned [22]. Similarly, Dror et al. found a positive correlation between the number of questions submitted or answered during the first visit and new user retention rates on Yahoo! Answers [15]. Additionally, Burke et al. reported that new Facebook users who demonstrated an initial inclination to contribute were more likely to continue posting in the future [7]. However, productive flyby users present an anomaly, as they invest substantial effort during their first visit yet still fail to return.

2.2 User Input Quality and Integrity

User input carries the risk of being untrustworthy or inconsistent, encompassing both natural noise—where users change ratings when asked to rerate movies after some time [12]—and malicious noise, such as a movie producer deliberately assigning high ratings to promote specific films [26]. Previous research has investigated user rating quality using two primary approaches: rerating consistency and simulation. Amatriain et al. asked users to rerate movies and addressed inconsistency by removing unreliable ratings [2]. Similarly, Cosley et al. and Nguyen et al. explored how user interfaces influence rating consistency in movie recommendation systems [12, 25]. Several studies proposed methods to assess and quantify rating quality and classify users' risk levels based on their inputs. For instance, Toledo et al. defined rating noise as the match between a rating and its predicted positive or negative direction [34], while O'Mahony et al. measured rating noise as the distance from predicted ratings and evaluated denoising algorithms through simulated attacks [26]. Additionally, Aktukmak et al. identified untrustworthy users as outliers in latent spaces and tested their attack detection algorithm using simulated random or average attacks [1]. However, no prior research has specifically analyzed the rating quality and integrity of productive flyby (PFB) users.

2.3 Gaps

As noted earlier, prior research on new users in recommender systems has largely overlooked PFB users, whose behaviors are counterintuitive and lack clear explanations. Furthermore, most studies addressing the risks of user input rely on simulated noise or define noise in terms of rating consistency. However, an essential question about the trustworthiness of ratings remains unaddressed: did users rate items they had not seen? This gap highlights the need to better understand PFB users' motivations, experiences, and the risks associated with their ratings – whether stemming from unseen items or inconsistency.

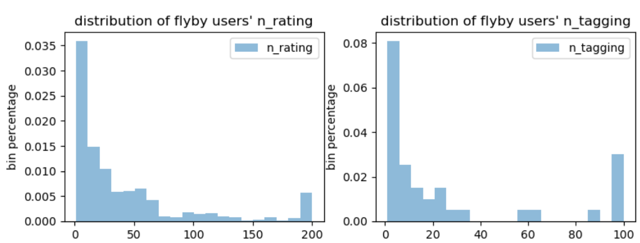

Figure 1: Distribution of the rated (left) and tagged (right) movies of flyby users on MovieLens. In both plots, the rightmost bins include n_rating ≥ 200 or n_tagging ≥ 100. Users who rated or tagged 0 movies were not counted.

3 Methods

Our methods are divided into three components: (1) identifying PFB users within the movie recommender system; (2) conducting survey and interview studies to explore PFB users' motivations, expectations, and reasons for leaving the platform; and (3) performing offline experiments to evaluate the impact of PFB users' ratings on regular users.

3.1 Define PFB users

We begin by explaining how we identified PFB users and why they may differ from other user groups. This study was conducted from April 2021 to April 2022, with **new users** defined as those who registered in our system after January 1, 2021, with a sign-up date at least one month prior to our initial analysis. **Flyby users** were defined as new users who never returned to the system after their initial session, which we specified as a 24-hour window following registration. For comparison, we selected a random sample of **regular users**, including new users who returned after their initial session and active users who logged in more than 12 times during 2020 and 2021.

To assess productivity, we analyzed activity logs from 980 flyby users who registered in the month preceding the study. We calculated the distributions of movies these users manually rated or tagged, as shown in Fig. 1. Based on this analysis, we defined **productive flyby users** as those whose number of ratings (*n_rating*) or tags (*n_tagging*) exceeded the 70th percentile of the overall distribution—specifically, more than 43 movie ratings or 20 movie taggings.

Using this definition, we categorized users into four groups, as summarized in Table 1, along with their statistical overview. We randomly selected 1,000 PFB users, 1,000 non-productive flyby users, and 2,000 regular users. Among the regular users, 1,000 were long-term active users who logged in more than 12 times during 2020 and 2021, while the remaining 1,000 were new users who registered in 2021 and returned after their initial session. Additionally, 642 of the 2,000 regular users were identified as productive.

Preliminary observations revealed notable differences between PFB users and other groups. PFB users demonstrated significantly higher activity levels during their first visit, with an average of 84 rated and 23 tagged movies—exceeding even those of productive regular users. Furthermore, 47.9% of PFB users rated unpopular movies during their initial session, and their average rating (3.7) was the lowest among all user groups.

3.2 Qualitative Understanding

We conducted a survey and an interview study to better understand PFB users' motivations, expectations, reasons of leaving, and rating integrity experience. This study was approved by our institutional IRB.

3.2.1 *Survey.*

The primary purpose of the survey was to ask PFB users to rerate some of the movies they had previously rated and identify potential rating integrity issues. The process for selecting survey movies was conducted in two stages. In Stage 1, four researchers performed a log analysis [16] on a random sample of 100 PFB users' rating histories. They identified rules for detecting "suspicious" ratings based on three criteria: movies with the lowest **popularity** (measured by the total number of ratings a movie received), movies rated with the shortest **time gap** following the user's prior actions, and movies with the highest variance between the user's rating and the predicted score.

In Stage 2, an algorithm was developed using the criteria established in Stage 1 to construct personalized surveys for each user. Each survey included 9-12 movies from the user's rating history, consisting of 2-3 movies from each of the three categories identified as potentially suspicious, along with 2 randomly selected movies. The survey asked the following questions:

User Group	First Visit Median of (# Movies Rated)	First Visit Median of (# Movies Tagged)	First Visit % Unpopular Movies Rated	All Visits Avg Rating
Productive Flyby Users (n=1000)	84	23	47.9%	3.7
Non-Productive Flyby Users (n=1000)	0	3	10.3%	4.2
All Regular Users (n=2000)	24	2	25.6%	4.1
Productive Regular Users (n=642)	76	4	41.3%	4.0

Table 1: The summary statistics for productive and non-productive regular or flyby users. The statistics on tagged movies exclude the users who did not use the tag feature. Unpopular movies are the movies with fewer than 300 ratings (the 90 percentile of our movie popularity, since our top 10% popular movies get 95% ratings). Our rating system is scaled from 0.5 to 5 stars with a half star interval.

(1) *Why did you come to MovieLens? What were you hoping to find?*
(2) *Why did you invest so much time in the site, and why have you not come back?*
(3) *Have you seen the following movies? If so, do you like them? Please select from the following choices: Not Seen / Disliked (rating <=2) / Neutral (rating 2.5 - 3.5) / Liked (rating >=4).*

The survey was distributed in four rounds. Initially, we emailed the survey to 30 PFB users, but received no responses. To improve the response rate, we introduced a $10 Amazon gift card incentive in the subsequent three rounds. Ultimately, the survey was sent to 449 PFB users who registered between November 1, 2021, and February 20, 2022, resulting in 41 responses. To establish a baseline for rating integrity, we distributed the same survey – including the identified "suspicious" rated movies—to 270 regular users, obtaining 20 responses.

3.2.2 Interview. To gain deeper insights into PFB users' motivations, expectations, reasons for leaving, and potential rating integrity issues, we conducted interviews with 11 survey respondents on a rolling basis. Recruitment ceased once no new patterns emerged from participants' input, indicating data saturation. Each interview lasted approximately 30 minutes; six were conducted via Zoom in video format, while the remaining five were text-based chats on Google Hangout. Participants were compensated with a $25 Amazon gift card upon completing the interview. A semi-structured interview format was employed, with questions organized around five main themes:

(1) *How did you find MovieLens? What motivates you to rate so many movies?*

Risk Level	Definition
Match (risk=0)	survey rating matches their original rating
Small Mismatch (risk=1)	survey rating is 0.5 ~ 1 above or below their original rating
Large Mismatch (risk=2)	survey rating is 1 star above or below their original rating.
Unseen (risk=3)	user reported that they had not seen this movie

Table 2: Rating Integrity Categories

(2) *You indicated that you did not watch <movie>, but you rated it on MovieLens. Why did you rate an unseen movie? How did you decide what rating to input? How did you hope the system to react?*
(3) *You gave <movie> x-star in the survey but y-star in the system. What might cause this difference? What is your rating criteria?*
(4) *Could you share your experience with tagging and wishlist features on our platform, if you have used them?*
(5) *Could you share some reasons for leaving the platform?*

We conducted an inductive thematic analysis [6] to identify common patterns in the interview data and responses to survey questions 1 and 2. Four researchers independently coded the interview transcripts, after which they collaborated to group and refine the identified clusters. In cases of disagreement, the researcher who conducted the corresponding interview provided additional context to clarify the participant's input. The team then discussed and voted on a final decision until all disagreements were resolved.

4 Results

For a quantitative overview, we summarized the statistics of all interviewed participants (P1 to P11) in Table 3. The **rating integrity risk** was defined based on its correlation with each participant's response to survey question 3, as detailed in Table 2. For each participant, their overall rating integrity risk level was determined by the highest level of risk observed in their responses.

4.1 Motivations for Coming

We identified five main motivations from survey and interview data.

4.1.1 Get Personalized Recommendations. In total, 17 out of 41 PFB users indicated in the survey that they came to the platform seeking **personalized** recommendations. Unlike passive recommendations from other websites, they actively contributed their ratings to receive suggestions tailored to their tastes. P6 explained this motivation, stating: *"I would spend most of my evenings scrolling through Netflix trying to find something to watch. And then realizing that I'd spent the whole evening and hadn't actually found any. So I decided I would try and find like a personalized recommendation system."* Similarly, P11 praised the platform's ability to go beyond popularity-based recommendations: *"I find it better than just a website that [shows] top 100 action movies. And the ability to give my own ratings and get suggestions."* This sentiment was echoed by P19, who shared: *"I hoped to find a site that helped me find movies that I might like based on movies I had rated."*

	n_rating	avg_rating	median of rated movie popularity		n_rating	avg_rating	median of rated movie popularity
P1 (risk=3)	256	3.955	9418.5	P7 (risk=2)	52	4.038	27275
P2 (risk=3)	232	3.537	20038	P8 (risk=2)	68	3.772	30008.5
P3 (risk=3)	171	4.719	7839	P9 (risk=1)	163	3.525	13125
P4 (risk=3)	95	3.363	18907	P10 (risk=0)	22	3.886	33952.5
P5 (risk=2)	125	4.296	69	P11 (risk=0)	45	3.933	15635
P6 (risk=2)	87	4.069	9247				

Table 3: Summary statistics of participants P1 to P11. *n_rating* **refers to absolute number of ratings;** *avg_rating* **indicates the mean of all ratings the user has (min: 1, max: 5); movie popularity is the count of all ratings on a single movie.**

Motivations	# users rated unseen movies / # of total users	Reasons for Leaving	User Group Description		# users rated unseen movies / # of total users
Get Personalized Recommendations	7/17	Never plan to stay	Come for class and research, usually one-shot usage		1/15
Class and Research	1/15	Failed Expectations	Dislike recommended movies	0/5	4/14
Find New and Surprising Movies	3/7		Prefer other movie recommendation websites, such as those with better UIs	1/5	
Find Similar Movies	2/5		Unsurprising recommendations	1/2	
Store or Share Ratings	1/2		Inaccurate prediction (low score on movies they like)	1/1	
			Lack of explanation on recommendations	1/1	
		Lack of Habits	Forget about this platform, nothing particular they dislike		6/13

Table 4: Summary of user motivations and leaving reasons based on the survey and interview responses. Each user can belong to multiple categories.

4.1.2 Class and Research. The second most common pattern involved 15 users who joined the platform for academic or research purposes. For example, some students were asked to "access and test a recommender" by their professor (P37).

4.1.3 Find Something New. Seven users highlighted the importance of discovering new movies, rather than being recommended options they had already encountered on other platforms. P11 expressed this sentiment, saying, "I thought 95 (ratings) provided more information, and it would be able to make better suggestions about what I don't know and might find interesting." Similarly, P14 emphasized the desire for novelty: "I needed a platform that would recommend movies based on my tastes, not what I've seen before."

4.1.4 Find Similar Movies. Five users came for more than pure personalization — they had specific movies in mind and hoped to discover similar ones. For instance, P9 shared, *"I was like mainly looking for movies similar to the movie Shutter Land"*. These users were particularly interested in finding recommendations based on certain attributes, such as community-contributed tags. As P2 explained, *"I'm interested in finding a similar movie based on the tag system"*.

4.1.5 Store or Share Ratings. In contrast to the users who sought new movies, some users expressed a preference for the system to recommend films they had already watched, even if they disliked them. As P5 stated, *"I still like the system to show these movies (watched and disliked), and I could input my ratings to them."*. Additionally, displaying movies the user might have already seen could help build trust with the system. As P11 explained, *"(Showing) you probably get a high probability watch these movies — I think that would say to me that the algorithm works well on learning my preferences."*

4.2 Expectations for High-Investment

In addition to the motivations driving PFB users to the platform, we hypothesize that these users hold additional expectations, especially considering their significant investment during their initial interaction. Through our analysis, we identified and categorized six primary expectations, which are detailed below.

4.2.1 Immediate Feedback. Some users expressed an expectation for recommendations to update instantly after they added a rating. For example, P9 remarked, *"I assumed it was automatic — as soon as I rate the movie, it gets factored in."* However, our recommender system primarily relies on collaborative filtering (CF) algorithms, specifically item-based CF [20] and singular value decomposition (SVD) [17], which generate recommendations on a daily basis. Upon learning about this delay, P9 suggested providing users with a notification to set appropriate expectations: *"So if there's a delay that you'd like to take, I did not know that. It should be, like, a note about that on the page."*

4.2.2 Explanation to Gain Trust. Users highlighted the importance of explaining how recommendations were generated to build trust with the system. For instance, P2 stated, *"You need to know why the movie was given this predicted rating, like because these people* *have similar scores to me — the transparency (of the algorithm)."* Providing clear and transparent explanations of the recommendation process can help users better understand and trust the system's functionality.

4.2.3 Recommend Based on Attributes. Users expected recommendation algorithms to recognize that their ratings often reflected specific attributes of the movies. Consequently, they hoped for future recommendations to incorporate this understanding. Commonly mentioned attributes included genres, scenes, music, and actors. For instance, P10 explained, *"Comedy elements in darker-toned shows are a big draw for me, and I'd like to see similar things recommended."* Similarly, P8 shared, *"Things are moving too slow (in this movie), but I'm a huge huge fan of Quentin Tarantino."* P9 even proposed a novel feature, suggesting the platform allow users to rate specific scenes: *"If you could do that (scene clip), then you could say okay, this is what this person likes, this is what makes him like this movie."*

4.2.4 Group Recommendation. Some users revealed that their extensive use of ratings and tags resulted from combining their own data with input from friends. They often used the system collaboratively to find movies for group viewing. For example, P7 explained, *"I wanted to get new movie recommendations that all my friends would like. For this reason, I aggregated all of their reviews and input them into the site."* However, as the platform was not designed for group recommendations, this approach led to inconsistencies in their profiles. Consequently, users like P7 were unable to receive personalized recommendations tailored to their individual preferences: *"When I compared two different ratings (from mine and my friends), they might seem good if you looked at them individually, but not make sense when you compare them."*

4.2.5 Follow A Real Person. Some users, like P2, expressed a preference for trusting recommendations from real people, such as well-known movie critics or friends they know personally: *"Typically I use Letterboxd friends' ratings and sometimes film critic reviews to find film reviewers (for recommendations)."*

4.2.6 Just Look Around. Some users, like P5, indicated that their extensive rating and tagging activity stemmed from being "a huge movie person." In contrast, P3 shared that they were simply passing the time by rating movies on their phone, not realizing how many ratings they had submitted in the process.

4.3 Reasons for Leaving

We clustered and summarized the main reasons of why PFB users left the system after their initial effort (see Table 4).

4.3.1 Never Plan to Stay. The first group, categorized as *Class and Research* users, had no intention of staying on the platform beyond completing their assignments or research tasks. Within this group, only 1 out of 15 users rated movies they had not seen.

4.3.2 Failed Expectations. The second subgroup left the platform due to unmet expectations. Among this group, rating integrity issues were more prevalent, with 4 out of 14 users having rated movies they had not seen.

4.3.3 Lack of Habits. The third subgroup is particularly interesting. These users stated they simply forgot about the website, without any major disappointments. Some attributed their departure to a lack of a movie rating habit: *"I haven't been back, maybe because of, like the need for a prompt or a reminder, did you watch these movies. (P5)"* However, we observed a challenge in designing such reminders, as many users didn't add movies they planned to watch to their wishlists or even clicked on them. Instead, they watched them immediately: *"If I see one movie that I like I would try and go immediately and watch that one movie. (P11)"* This group had the highest occurrence of rating unseen movies (6/13).

These varying user behaviors highlight the need to study user inaction [42]. Given their different motivations and reasons for leaving, we speculate that these three subgroups may exhibit different data quality. In the next section, we examine their ratings' risk of being for unseen movies or inconsistent.

4.4 Risk of Rating Integrity

From survey question 3, we found that **PFB users rated more unseen movies** compared to regular users. Table 5 highlights the basic rating integrity statistics of these groups. Specifically, 6% of PFB users' ratings were for unseen movies, double the 3% of regular users. Additionally, 27% of PFB users rated unseen movies, compared to only 15% of regular users.

Interestingly, **PFB users who joined for class or research purposes rated fewer unseen movies.** Only 1 out of 15 users in this group rated unseen movies, likely due to project requirements. This suggests that platforms can potentially maintain data integrity with a mix of regular and academic users.

In contrast, **PFB users who left due to a lack of movie rating habits rated more unseen movies** (see Table 4). These users often engaged with the platform to pass time and exhibited less concern for rating accuracy. For instance, P1, who rated over 100 movies, remarked: *"I think I just had one of those days when the most interesting thing in the world was to rate movies. I am surprised that it was that high."* They also acknowledged instances of accidental inputs: *"I probably just accidentally clicked."* Similarly, P3 shared: *"I just sat on my phone going through movie after movie."*

4.5 Reasons Behind the Risk

After observing the randomness and high frequency of rating unseen movies among PFB users, we sought to better understand the underlying reasons for such behavior. Beyond less deliberate actions, such as misinputs or the potential influence of their motivations, PFB users offered additional insights into when and how they intentionally rated unseen movies or provided inconsistent ratings.

4.5.1 Gain Control of Algorithms. Each user applied a different personal standard when rating movies. Frequently, they could predict they would not enjoy a movie based solely on its description or premise. In such cases, users rated these movies with a low score to signal the algorithm to avoid recommending similar content in the future: *"I know those are movies that I do not like. I was probably trying to get that type out of the running for recommendations. (P2)."* These ratings were often polarized as a means of "controlling the

Rating Subgroup	PFB users # users = 41 # ratings = 413 (# ratings, % ratings)	Regular users # users = 20 # ratings = 200 (# ratings, % ratings)	User Subgroup	PFB users # users = 41 (# users, % users)	Regular users # users = 20 (# users, % users)
Match (risk=0)	309 (75%)	158 (79%)	**max(risk) = 0**	6 (15%)	4 (20%)
Small Mismatch (risk=1)	42 (10%)	23 (12%)	**max(risk) = 1**	9 (22%)	6 (30%)
Large Mismatch (risk=2)	38 (9%)	13 (6%)	**max(risk) = 2**	15 (36%)	7 (35%)
Unseen (risk=3)	24 (6%)	6 (3%)	**max(risk) = 3**	11 (27%)	3 (15%)

Table 5: Statistics on rating integrity risk of PFB users and regular users based on responses from survey question 3.

algorithm," resulting in scores significantly lower than the system's predicted ratings. Such patterns merit further investigation.

While dislike could often be presumed without watching a movie, the opposite—liking a movie—typically required viewing it first. As P1 shared: *"This is something that I imagine will be so bad. I don't want to watch it. Conversely, if you like the kids' movies, you watch them."* Polarized ratings also occurred with medium-level movies, particularly when users sought to adjust the types of recommendations they received. For example, P2 deliberately gave a medium-level movie a 1-star rating to "fine-tune" the recommendations: *"I put it, because I wanted to fine-tune the recommendations."*

However, the recommender system, powered by algorithms such as K-nearest neighbor collaborative filtering [20] and matrix factorization [17], is not inherently polarization-aware [5]. These algorithms operate under the assumption that users rate only movies they have actually watched, which may limit their ability to accommodate such user-driven interventions effectively.

4.5.2 Changes of Contexts. Users gave different ratings under different moods, other movies displayed on the same list, and the existence of system-predicted ratings. P1 rated one movie lowly in the survey because of their mood: *"I was just not in the mood (when taking the survey) because now that you mentioned it again, I liked that movie"*. P6 gave different ratings because the set of movies changed: *"When I see a different set of films, I kinda think of them differently."* P8 suggested the platform to make predicted score display optional on movie posters because it interfered with their judgment: *"The estimate, I think it's kind of affects me like as a person. I hope it was not visible at first, or if there was a setting to make, like don't show the recommendations until I rated."* In practice, MovieLens was not running any of the mentioned mechanisms to guarantee the rating consistency.

4.5.3 Unremarkable Mediocre Movies. Mediocre movies posed a challenge for users to remember and rate consistently compared to movies they clearly liked or disliked. Often, users would neither search for nor rate such movies unless prompted by the recommendation system. As P9 explained: *"To me, that was like a really forgettable movie. So I wouldn't have rated it unless, see, if it came up (with recommender)."*

4.5.4 Other Movie Attributes. As discussed in Section 4.2.3, users often rated movies based on different attributes such as actors, stories, or music. This practice led to varying ratings depending on the specific aspects they focused on. However, the collaborative filtering (CF) algorithms behind the platform were not designed to distinguish which attributes users prioritized in their ratings.

Participant feedback revealed a disconnect between user assumptions and algorithmic design. Many PFB users operated under incorrect assumptions — for instance, that the algorithms would account for their intentions when they rated unseen movies to manipulate recommendations. While it is understandable that users may struggle to maintain rating consistency due to changing contexts or their perception of mediocre movies, CF and similar recommender algorithms generally lack the ability to handle ratings influenced by detailed attributes.

Overall, PFB users exhibited unique behaviors, including rating more unseen movies than regular users, giving polarized ratings, and holding unrealistic expectations about how the system operates. These behaviors introduce potential challenges for the recommendation algorithms. As the next step, we conduct an offline evaluation to assess the extent to which PFB users with such assumptions affect the performance of the algorithms for regular users.

5 Offline Rating Risk Evaluation

To test how ratings from PFB user affected the recommendation performance for other regular users in our system, we conducted offline evaluations in three folds:

(1) **Total Effects**: How can all PFB users affect other regular users?
(2) **User Subgroup Effects**: How can specific subgroups of PFB users affect other regular users?
(3) **Rating Subgroup Effects**: How can specific rating subgroups of PFB users affect other regular users?

To evaluate the effects of PFB users' behaviors on recommendation algorithms, we created a dataset combining all ratings from 1,000 PFB users identified in the system with those from 2,000 regular users (see dataset details in Table 6). Both groups contributed roughly the same number of ratings (see Table 1). We constructed a baseline dataset containing all ratings from both PFB and regular users. In addition, we created three experimental datasets to assess different effects:

(1) **Total Effect Training Set**: Ratings from all PFB users were removed, leaving only the ratings from regular users.
(2) **User Subgroup Effect Set**: Ratings from high-risk PFB users were removed. High-risk users were identified based on patterns observed in their rating

logs, interviews, and survey responses (see Section 5.2 for details).

(3) **Rating Subgroup Effect Set**: High-risk ratings from PFB users were removed. These ratings were flagged based on detailed observations of rating attributes and their temporal distribution (see Section 5.3).

We tested three CF algorithms, two of which (Item-based CF and SVD) are currently implemented on MovieLens: 1) User-based K-nearest Neighbor (n_neighbor=10); 2) Item-based K-nearest Neighbor (n_neighbor=10) [20]; 3) FunkSVD (n_factor=20) [17]. For the train-test split, we applied 10-fold cross-validation with a 70-30 train-test ratio across the baseline dataset and the three experimental datasets. Our evaluation metrics include the average *RMSE* and *precision@8* (as we show top 8 movies in each recommendation carousel on the homepage; the threshold for relevance is defined as recommended item rating ≥ 3.5 stars), and the associated Cohen effect size [31] between baseline performance and each experiment groups base on results of paired t-tests. To ensure statistical robustness, we applied Bonferroni correction [36] to control for Type I error in multiple hypothesis testing.

5.1 Total Effects

Our analysis reveals that **PFB users as a whole do not impact** recommendation performance for regular users (see Table 6, Total Effects row). When PFB users are excluded from the training set, both user-based and item-based CF algorithms exhibit significantly higher RMSEs (0.777 and 0.794, respectively) compared to the baseline error rate. Additionally, the precision@8 score for SVD (0.851) drops significantly when trained only on regular users' data, relative to the baseline dataset that includes both regular and PFB users' ratings.

These findings align with our survey results, which indicate that the majority of PFB users' ratings (94%) correspond to movies they have watched (see Table 5). However, further qualitative and quantitative analysis suggests that certain subgroups of PFB users demonstrate salient rating integrity issues. These challenges will be explored in greater detail in the subsequent sections.

5.2 User Subgroup Effects

As previously discussed, we hypothesize that the motivations and reasons for leaving the platform among PFB users may correlate with their rating integrity risks. However, the subjective information required to validate this hypothesis is absent for the majority of users in our dataset. To address this limitation, we conducted a data-driven exploration of various rating features to uncover patterns indicative of rating integrity risks. By combining insights from 41 survey responses with log data analysis, we identified the following prominent patterns among users with the highest rating integrity risk:

(1) Three out of six PFB users whose **rating_count ≥ 200** had rated unseen movies;
(2) Three out of four PFB users whose **avg(rating - predicted_rating) < -0.25** had rated unseen movies;
(3) Two out of two users whose **avg_rating < 3** had rated unseen movies;
(4) Two out of three users whose **median(rated_movie_popularity) < 1000** had rated unseen movies

Relying on the discovered patterns, we further investigated and found that the four users with two or more of these patterns had rated unseen movies. This number is 2 out of 6 for users with one pattern and 5 out of 30 for users with zero patterns. To quantify user subgroup effects, we conducted an experiment to test the recommendation performance by excluding the subgroup of around 27,000 ratings from 122 high-risk PFB users who have **two or more** of these patterns from the training dataset.

The high-risk PFB users significantly increased the error rate of the SVD algorithm on regular users. When removing the 26,000 ratings of the 122 high-risk users, the RMSE value of SVD decreased significantly to 0.798 compared to the baseline trained with the 273k ratings of all users (0.803).

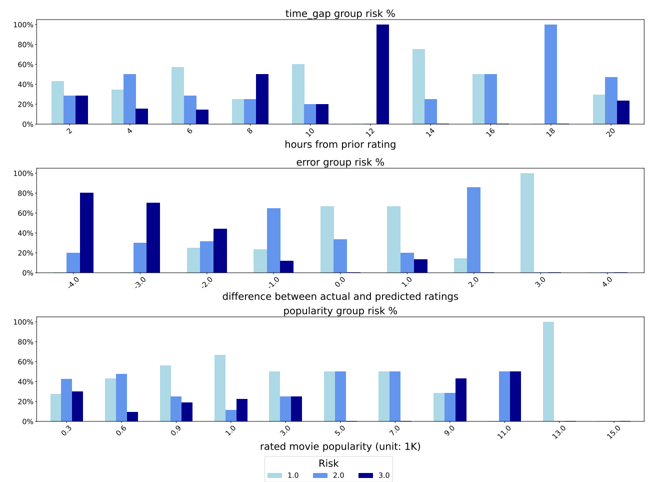

Figure 2: The distribution of the PFB user rating integrity risk under three patterns. Up: rating time_gap; Mid: rating error (*actual rating - predicted rating*); Bottom: rated movie popularity (base unit as 1000). Ratings with risk=0 were not displayed. See Table 2 for the definition of rating integrity risk categories.

5.3 Rating Subgroup Effects

Though excluding high-risk PFB users significantly decreased the recommendation error rate for regular users, we suspect that not all of PFB users' ratings have a negative impact. To isolate more specific high-risk ratings, we plot the distribution of PFB users' rating integrity categories based on observed patterns in three dimensions:

(1) Rating risk distribution in terms of 2-hour time intervals.
(2) Rating risk distribution in terms of error group, where $error = actual\_rating - predicted\_rating$.
(3) Rating risk distribution in terms of movie popularity (divided by 1000 for easier numeric representation).

Experiment	Dataset	*Recommendation Performance for Regular Users*					
		User-based CF		**Item-based CF**		**SVD**	
		RMSE	Precision@8	RMSE	Precision@8	RMSE	Precision@8
Baseline	2000 Regular Users (136k ratings) + 1000 PFB Users (137k ratings)	0.773	0.844	0.781	0.849	0.803	0.856
Total Effects (Exclude all PFB users)	2000 Regular User (136k ratings)	**0.777*** **(0.000, 0.746)**	0.842	**0.794*** **(0.000, 2.275)**	0.846	0.803	**0.851*** **(0.006, 1.262)**
User Subgroup Effects (Exclude high-risk PFB users)	2000 Regular User (136k ratings) + 878 Low-risk PFB Users (110k ratings)	0.772	0.844	0.780	0.849	**0.798*** **(0.004, 0.845)**	0.855
Rating Subgroup Effects (Exclude high-risk ratings of PFB users)	2000 Regular User (136k ratings) + Low-risk Ratings from 1000 PFB Users (109k ratings)	**0.770*** **(0.018, 0.276)**	0.844	**0.777*** **(0.002, 0.609)**	0.850	**0.798*** **(0.006, 0.825)**	0.857

Table 6: Summary of the recommendation algorithms' average performance for regular users when trained by different data. The groups that perform significantly better (blue) or worse (orange) than the baseline are indicated by *(adjusted p-value [36], cohen size [31]).

From Figure 2, we observed the following: 1) no clear pattern was associated with the time gaps when PFB users contributed their ratings; 2) the lower the error, the higher the rating integrity risk, especially when $error \leq 2$; 3) less popular movies have a higher rating integrity risk than more popular ones. Based on this observation, we excluded 27 thousand **"high-risk ratings"** from PFB users, which were either 2 stars lower than the predicted ratings or for unpopular movies ($popularity < 300$). We then followed the same training procedures to quantify the rating subgroup effects.

The high-risk ratings from PFB users significantly increased the prediction error for all three CF methods on regular users. Without them, the RMSE value for SVD slightly dropped from 0.803 (baseline) to 0.798; for Item-based CF, it dropped from 0.781 to 0.777; and for User-based CF, the error rate decreased from 0.773 to 0.770. Precision@8 was not significantly affected in any of the three algorithms.

6 Discussion

Overall, our findings regarding PFB user behaviors and rating integrity challenges differ from many prior studies on new users. While numerous studies interpreted new user investment on platforms as an indicator of satisfaction or intention to stay [7, 15, 22], our qualitative analysis suggests that productivity can also stem from other reasons, such as the desire to control the algorithms for immediate feedback, contributions from a group of people, or simply killing time or browsing. Based on these varying motivations and expectations, rather than retaining PFB users [21], we demonstrated the negative impact on recommendation performance for regular users. Specifically, we showed how training CF-based algorithms with high-risk PFB users and their high-risk ratings negatively affected regular users' recommendation experience, using a series of offline evaluations. In the following sections, we will expand the discussion, focusing on the three research questions we introduced earlier in the paper.

6.1 PFB Users' Motivation, Expectation, and Reasons for Leaving (RQ1)

PFB users expressed various motivations for visiting MovieLens. While many indicated they came for personalized or new movie recommendations (aligning with the goal of the recommender), we identified a challenge in addressing two types of user needs: those seeking surprise recommendations and those seeking similar or even previously watched movie recommendations. A key design implication from this finding is the need to clearly inform users about why certain recommendations are displayed. Is it because the system believes they might have already watched those movies and needs their ratings for better personalization? Or is it because the model predicts they haven't watched them yet and might enjoy the recommended movies? To address these needs, more explainable interfaces [18, 33, 41] and interactive designs [32, 35] are essential.

Most of the expectations shared by PFB users did not align with the design of MovieLens. The two main features we currently support are: 1) building custom profiles for personalized recommendations, and 2) finding movies similar to those a user has shown interest in. We assumed that regular users were satisfied with these two features for their long-term engagement with the platform. However, this reveals a gap in the transparent communication that recommender systems should establish with new users. While it can be technically challenging to support personalized recommendations across different contexts, PFB users expressed a desire for recommendations based on specific movie attributes, such as plot or clips. In this case, we should advise users to explore other platforms powered by computer vision [24] or multi-modality recommendation models [38].

Admittedly, PFB users had diverse expectations, some of which may conflict with each other or even contradict the interests of regular users. For instance, regular users of MovieLens likely appreciate the platform's ease of use and its simple concept—rate movies to receive predicted ratings. However, if we were to implement design changes such as group recommendations (a feature we previously added and later removed due to low usage by regular users), hiding predicted ratings, or allowing users to rate movies based on specific attributes, the system's complexity would increase, potentially diminishing its appeal to regular users.

We summarize the actionable implications based on our findings regarding PFB users' motivations and expectations: 1) Recommender system practitioners should regularly assess user expectations and enhance the platform based on feasible improvements that align with the system's core concepts; 2) Clear communication through explainable interfaces or user interactions is essential to inform users about the system's capabilities. This approach can help prevent PFB users from investing excessive time during their first session with unrealistic expectations. For example, prior work by Cascaes Cardoso has suggested using a "purpose statement" during the onboarding process to clarify what the platform can and cannot do [9].

6.2 Rating Integrity Risk and Systematic Impact of PFB Users (RQ2 & RQ3)

While 27% of PFB users reported rating unseen movies in our survey and interviews, PFB users, as a whole, positively contributed to the recommendation accuracy for regular users. Admittedly, the ratio of PFB users to regular users in a system can influence prediction shifts. As more data generally benefits generalizability, future recommendation algorithm designers should carefully explore how to incorporate low-risk PFB users and their ratings to enhance the overall recommendation performance of the system.

Researchers have examined various attack and defense strategies for adversarial recommender systems [13], but few have linked malicious data detection to users' contribution patterns. In some cases, high-risk PFB user subgroups and low-quality ratings increased the recommendation error rate for regular users. While we observed high-risk usage patterns in the movie domain, we believe this knowledge can be applied to recommender systems in other domains as well. For instance, if a PFB user consistently gives low ratings on an e-commerce platform or contributes an unusually high number of ratings for unpopular songs on a music recommendation platform, the system should reassess their input and establish a mechanism for checking training data quality. As different platforms have unique user behavior patterns and rating scales, and item distributions cannot be universally applied, we suggest that future researchers consider modeling users based on their productivity and other personal attributes.

7 Limitations and Future Implications

We acknowledge several limitations of this study. First, it is a single-domain case study, with findings rooted in the specific context of movie recommendations. In other domains, user inputs might follow different economic models [19]. For example, contributing 200 ratings or interaction data points may be considered typical for new users on platforms like short-video recommenders. Second, the offline evaluation relies on a relatively small dataset. Other platforms may have more or fewer PFB users, potentially affecting regular users in different ways. Despite these limitations, we believe our qualitative findings and evaluation of rating integrity can provide valuable insights for practitioners in other domains or platforms, helping them better understand and model PFB user behaviors and the risks associated with their ratings.

Future work can be divided into two directions. In addressing the rating integrity challenge, offline evaluations could be extended to analyze larger datasets, encompassing more PFB and regular users, and examining subgroup impacts between the two groups more comprehensively. Another direction involves exploring new interface or algorithmic designs that balance the need to meet PFB users' reasonable expectations while continuing to serve regular users effectively.

8 Conclusion

In this study, we investigated the motivations, expectations, and reasons of leaving for productive flyby (PFB) users on a movie recommender platform. Through surveys with 41 PFB users and in-depth interviews with 11 of them, we uncovered various factors driving their behaviors. Notably, 27% of PFB users reported rating movies they had not watched. To further understand the impact of such behaviors, we conducted offline evaluations, identifying rating integrity risk patterns and segregating specific user and rating subgroups from the recommendation algorithm training process. Our findings revealed that high-risk PFB user subgroups and their ratings can negatively affect the recommendation performance for regular users on the platform. Based on these insights, we proposed design implications to enhance the understanding of user motivations and expectations, as well as strategies for adversarial checks to mitigate rating integrity risks. We hope this case study raises awareness of this unique user group within the broader human-centered recommender system research community.

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